Computational Astrophysics 2023/2024
Physics of Data

Part-1: *N*-body simulations
Submission deadline: 23/12/2023

Assignment 1: Better Together!

"It is the long history of humankind (and animal kind, too) that those who learned to collaborate and improvise most effectively have prevailed."

C. Darwin

Learning aims: Through the solution of this exercise, you will set-up the gitlab repository of your project and connect it with your local working machines using Git. You will also make experience with the basic tools and commands to manage your Git repository.

Tasks:

A. **Let's start:** Fork the project Gitlab repository from https://gitlab.com/iogiul/pod_compastro23.git. Then, invite me to your Gitlab repository as a member with role "developer". My Gitlab username is @iogiul. It is up to you to decide if you want to have a public or private repository. *(My addition to the repository is enough to complete the task. In the assignment submission, just report the name and link of the repository).*

B. **Don't forget your parents:** Add the original Gitlab repository as a remote and call it **ca23_remote**. Fetch the data from this remote. Get the date and the hashcode of the last commit of the remote. Add this information in your main branch in a file called **"assignment1_taskB.txt"**, commit and push this file to your remote. *(In the submission report the date and hashcode of the commit associated to the creation of the file assignment1_taskB.txt.)*

C. **Branch creation:** Create a new branch called **assignment1_taskC,** push it to your remote. Then, each member of the group has to create a file with name **taskC_gitlabusername.txt** where *gitlabusername* is the member username in gitlab. In the file put your real name. Finally,

each member has to commit and push this file using her/his gitlab account. *(In the submission report the username, the date date and hashcode of the commits made by all the members).*

D. **Let's merge:** Synchronise the content of the **main** and **assignment1_taskC** branch using the method/command your prefer (e.g. merger, rebase, cherry pick). *(In the submission briefly describe the method you used and report the date and hashcode of the commits produced by the merger).*

E. **Oh no!:** Modify the file **"assignment1_taskB.txt"** both in the branch **main** and **assignment1_taskC** in a way to create a merger conflict. Then merge the content of the branch **assignment1_taskC** in the **main** branch. Solve the conflicts, commit and push the final merged product. *(In the submission briefly describe how you create the merger conflict and how you solved it. Report the date and hashcode of the commits produced after the merger conflict solution).*

F. **Oh no!-2:** Repeat the task E, but this time using the **merger request** tool in Gitlab. Use Gitlab also to fix the conflict and conclude the merger. *(In the submission send me the link to the Gitlab merge request. Report the date and hashcode of the commits produced after the merger conflict solution).*

Challenge (not mandatory):

In Astrophysics we use a lot acronyms (personally I don't like this that much….) for defining the name of both software and instruments/ observational surveys. Few examples:

- SEVN (Stellar Evolution *N*-body), a code to evolve star and binary (you will use it)
- WEAVE (WHT Enhanced Area Velocity Explorer), a spectroscopic survey
- MANGA (Mapping Nearby Galaxies at APO), another spectroscopic survey
- GRAPE (GRAvity PipE), a special purpose hardware for *N*-body simulations

and many others…..

Do you want to "taste" the excitement of an astrophysics finding the perfect acronym (Yes, I am sarcastic..)? I challenge you on finding an acronym for

the name of your forked repository.. We (I and the teaching assistants) will elect the best three (No you will not have more “exam points” for this..).